



Inspection Report

University of Maryland Baltimore
10 S PINE ST. RM G-100, MSTF BLDG
BALTIMORE, MD 21201

Customer ID: **89**
Certificate: **51-R-0018**
Site: 001
VETERINARY RESOURCES

Type: ROUTINE INSPECTION
Date: 03-FEB-2026

2.31(c)(7)

Repeat

Institutional Animal Care and Use Committee (IACUC).

In response to an internal complaint representatives from the IACUC oversaw procedures conducted for Protocol #RA006 on August 20th and August 27th-28th. The ACUC observed the research staff care and use of 5 study rabbits. During their observations, the IACUC noted numerous deviations from the procedures outlined in the approved protocol. The deviations are as follows:

1. Syringes and catheters were reused at least twice per animal. The protocol indicates single use of syringes and catheters.
2. A cat bag was used to restrain the rabbits. The protocol describes the use of the "burrito" method for rabbit restraint.
3. No record of isoflurane induction. The protocol indicates the rabbits will be induced with isoflurane prior to the administration of ketamine/xylazine for anesthesia.
5. The ketamine/xylazine was administered intravenously as a bolus. The protocol indicates the drug combination should be administered intravenously by continuous infusion.
6. Research staff performed the resuscitation procedure while the rabbits were conscious. The protocol indicates the procedure should be performed under anesthesia.
7. Research staff did not consistently use aseptic techniques to clean the VAB ports. The staff acknowledged that the protocol indicates the VAB ports should be cleaned with 70% alcohol prior to syringe connection for blood collection and agent administration.

Numerous procedures were performed by the research staff in a manner not outlined in the protocol. The research staff continues to have protocol compliance issues following corrective actions implemented by the IACUC in April and August 2024.

Procedures in an IACUC approved protocol must be followed to ensure animals do not experience unnecessary pain or

Prepared By: GLORIA MCFADDEN
USDA, APHIS, Animal Care
Title: VETERINARY MEDICAL
OFFICER

Date:
12-MAR-2026

Received by Title: Facility Representative

Date:
12-MAR-2026



Inspection Report

distress and to ensure processes are conducted in a manner consistent with industry standards. Proposed activities for animal use that are approved by the IACUC should be followed or further appropriate review and approval would be needed in accordance with 2.31(d) of Subpart C. Correction: The IACUC assigned another principal investigator to Protocol #RA006 to implement a corrective action plan.

2.31(d)(1)(i)

Institutional Animal Care and Use Committee (IACUC).

After a discussion with lab members and procedural observations related to Protocol #RA006, the IACUC noted several instances of non-compliance that may have resulted in unnecessary pain and distress:

During IACUC observations of the hemorrhage procedure on August 20, 2025, two rabbits were anesthetized with Ketamine/Xylazine (IM). Both animals roused and regained consciousness approximately 30 minutes after the beginning of the procedure. Animals were righting themselves from a side lying position to sitting up and moving around in the rabbit bin. One animal attempted to hop out of the bin several times. Research staff were required to physically restrain the animals while still manually pulling blood for the hemorrhage. Animals appeared to intermittently lose and regain consciousness (becoming very active) as the hemorrhage procedure continued.

During observations on August 28, 2026, a rabbit was redosed 4 times to maintain sedation. The animal roused quickly and was alert after each dose wore off. Additionally, IACUC noticed the telemetry device readings did not change to indicate the rabbit was awakening. The monitoring parameters of the telemetry device is not effective in determining the rabbit is coming out of a plane of anesthesia.

Proposed activities must include procedures that will avoid or minimize discomfort, distress, and pain to animals. Correction: The IACUC required the research staff to be retrained to address these non-compliances. Staff received training on 09/2025 and 01/2026 for Protocol #RA006.

2.31(d)(1)(v)

Institutional Animal Care and Use Committee (IACUC).

1. The endpoint criteria in Protocol #RA006, relies on set parameters for Mean Arterial Pressure (MAP) that will remotely trigger an alert to the principal investigator if the readings for MAP reaches a certain percentage of the normal range (70mm Hg). However, conflicting low-pressure parameters are noted in the protocol - MAP less than 40mm Hg versus less than 35mm Hg (50% less than normal).

2. Protocol #RA006 indicates that once a trigger message from the telemetry device is received a member of the research staff will be present to determine if vital signs meet the endpoint criteria. Research staff indicated that the trigger messages is sent when the vital signs are in euthanasia range and they are allowed 30 minutes to arrive at the facility to evaluate and/ or euthanize the rabbit. Animals may be in pain or distress for up to an hour. The expectation as outlined in the protocol is that the rabbits would be euthanized when the endpoint criteria is met. The IACUC concluded that allowing the animal to be maintained longer than 30 minutes increases the likelihood that death would be the endpoint. The intent of the endpoint parameters is to intervene by euthanasia.

Prepared By: GLORIA MCFADDEN
USDA, APHIS, Animal Care
Title: VETERINARY MEDICAL
OFFICER

Date:
12-MAR-2026

Received by Title: Facility Representative

Date:
12-MAR-2026



Inspection Report

3. An IACUC review of telemetry data from Protocol #RA006 for one animal record showed the MAP spiked for a period of 15-20 minutes before rapidly declining. The animal died within " 4 minutes of the decline.

Additionally, during the USDA inspection, records were reviewed for three rabbits that were found dead. The telemetry data for two rabbits showed a sharp drop in MAP below the established MAP endpoint parameter and the animals died within 3 minutes of the sudden change. The MAP for one rabbit dropped below the established parameter continuously for a period of approximately 45 minutes before the animal died. The data suggests these rabbits may have experienced unrelieved pain or distress.

The MAP trigger parameter and the process for intervention is not effective in ensuring distress and pain is relieved in a timely manner. Proposed activities in which animals may experience severe or chronic distress or pain that cannot be relieved should be painlessly euthanized when endpoint criteria are met. Correction: The IACUC approved amendments to Protocol #RA006 to address this issue on December 17, 2025.

2.32(b)

Personnel qualifications.

2.32(b) During an IACUC oversight reviews of Protocol #RA006 in August 2025, it was noted that the research staff, including the trainers, were unfamiliar with the procedures as outlined in protocol. Research staff performing procedures asked the IACUC observers questions that indicated they did not know procedural details that were clearly stated in the protocol:

1. Research staff asked if the protocol indicated that rabbits should be sedated with xylazine/ketamine prior to euthanasia.
2. Research staff asked if Cathflo could be administered a second time after the waiting period or if the animal needed to be euthanized.

After discussions with consulting staff and the principal investigator, and the lab staff, the IACUC concluded that the persons providing training were unfamiliar with the details of the protocol and not knowledgeable of the processes and procedures as outlined in the protocol.

Also, during the IACUC review, only one lab member received the training established in 2024 as part of a corrective action plan for previous training deficiencies. Records indicate new lab members did not receive the same training.

Additionally, the IACUC noted that surgery and experimental records were incomplete with missing documentation of dates for procedures, anesthesia doses, blood pressure measurements. Some forms did not have signatures and dates indicating the record was reviewed as required as part of a corrective action plan in 2024. During the USDA records review, it was noted that prior to the IACUC observations, many records had very few notes documenting procedures performed on the rabbits.

The current research team for Protocol #RA006 were not familiar with the processes and procedures outlined in SOPs and the protocol to adequately perform their study responsibilities. Qualifications of research personnel must be reviewed with sufficient frequency to fulfill the research facilities responsibilities under this section and section 2.31. Correct by

Prepared By: GLORIA MCFADDEN
USDA, APHIS, Animal Care

Date:
12-MAR-2026

Title: VETERINARY MEDICAL
OFFICER

Received by Title: Facility Representative

Date:
12-MAR-2026



Inspection Report

training all new research staff and conduct training whenever there is an approved protocol amendment. NOTE: The current research team completed training in January 2026.

2.32(c)(1)(iv)

Personnel qualifications.

During the IACUC observations and discussions for Protocol #RA006, research staff members indicated they did not aseptically clean the VAB ports prior to connection for blood collection and agent administration. as outlined in the protocol. Additionally, the syringes and catheters were placed in the rabbit bin with feces and urine. Personnel must be trained in the importance of sterile techniques to prevent contamination and infection. Correction: The current research team completed training in January 2026.

2.32(c)(3)

Personnel qualifications.

August 20, 2025, the IACUC observer informed the research staff that the Protocol #RA006 allows for "top-off doses" of ketamine/xylazine to maintain unconsciousness during procedures. Research staff indicated they were not aware that the protocol has provisions to administer additional doses of ketamine/xylazine IM ("top-off" doses). On August 28, 2025, after reading the protocol, the research staff believed a maximum of 4 top-off doses were allowed. The protocol does not indicate a minimum or maximum number of doses.

During observations on August 27, 2025, the IACUC noticed that the anesthetic equipment was connected improperly and showed the research staff members the correct way to set up the equipment.

On August 28, 2025, the veterinarian noticed that the isoflurane level did not reach a maintenance level of anesthesia as outlined in the protocol and facilitated replacement with a properly working anesthetic machine.

Research staff were not knowledgeable of the proper administration of anesthesia. Continuous review and training of study processes and procedures is necessary to ensure animals are adequately anesthetized when study activities are performed. Correction: The current research team completed training in January 2026.

This inspection and exit interview were conducted with the facility representatives.

No study activities were conducted in 2026 to fully assess compliance. Further review will occur at the next inspection.

Prepared By: GLORIA MCFADDEN
USDA, APHIS, Animal Care
Title: VETERINARY MEDICAL
OFFICER

Date:
12-MAR-2026

Received by Title: Facility Representative

Date:
12-MAR-2026



Species Inspected

Cust No	Cert No	Site	Site Name	Inspection
89	51-R-0018	001	VETERINARY RESOURCES	03-FEB-2026

Count	Scientific Name	Common Name
000009	<i>Macaca mulatta</i>	RHESUS MACAQUE
000039	<i>Papio anubis</i>	OLIVE BABOON
000029	<i>Oryctolagus cuniculus</i>	DOMESTIC RABBIT / EUROPEAN RABBIT
000002	<i>Ovis aries aries</i>	SHEEP INCLUDING ALL DOMESTIC BREEDS
000003	<i>Sus scrofa domestica</i>	DOMESTIC PIG / POTBELLY PIG / MICRO PIG
000019	<i>Mustela putorius furo</i>	DOMESTIC FERRET
000005	<i>Cavia porcellus</i>	DOMESTIC GUINEA PIG
000106	Total	